

Based on your comfort level, you can select any one of the following tools.

- ChatterBot:** This has a Python library that makes it easy to generate automated responses for a user's input.
- NLTK:** This has a Python based open source natural language processing platform embedded with a huge set of methods and corpus packages.
- Scikit Learn:** This provides a range of supervised and unsupervised machine learning algorithms through a consistent interface in Python.

One can use a combination of different tools and libraries to design an efficient chatbot, as shown below in the code snippet of a simple response to a greeting:

```
greeting_input = ("hello", "hi", "greetings", "sup", "what's up", "hey",)
greeting_responses = ["hi", "hey", "**nods**", "hi there", "hello", "I am glad! You are talking to me"]
def greeting(sentence):
    """If user's input is a greeting, return a greeting response"""
    for word in sentence.split():
        if word.lower() in greeting_input:
            print random.choice(greeting_responses)
```

The following code snippet is for a dynamic response to a greeting, using Scikit-learn and NLTK:

```
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity
import nltk
import numpy as np
nltk.download('punkt') # one-time use only
nltk.download('wordnet') # one-time use only

sent_tokens = nltk.sent_tokenize(raw)# converts to list of sentences
word_tokens = nltk.word_tokenize(raw)# converts to list of words

flag=True
while(flag==True):
    user_response = input()
    user_response=user_response.lower()
    if(user_response!='bye'):
        if(user_response=='thanks' or user_response=='thank you'):
            flag=False
            print "BOT: You are welcome!!"
        else:
            if(greeting(user_response)!=None):
                print "BOT: "+greeting(user_response)
            else:
                sent_tokens.append(user_response)
```

```
word_tokens=word_tokens+nltk.word_tokenize(user_response)
final_words=list(set(word_tokens))
print "BOT: "
print response(user_response)
sent_tokens.remove(user_response)

else:

    flag=False
    print "BOT: Bye! See you soon.."
```

 **Note:** This snippet is just for purposes of understanding and more code may be needed to make it work.

END 

References

- [1] <https://www.analyticsvidhya.com/>
- [2] <https://en.wikipedia.org/wiki/Chatbot>

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-Thomas
Babington Macaulay

